

SIMATS ENGINEERING ASSESSMENT TOOL 3

NAME:C.Venkata Deepan
COURSE NAME: CSA1607

REG.NO: 192425022
COURSE CODE: Data Warehousing and Data Mining

COURSE OUTCOME COVERED

CO4: Examine the applicability of clustering algorithms in real-world scenarios, presenting findings through clear reports with effective visualizations.
Assessment Tool Weightage: 20%

QUESTIONS: 05

Total Marks:50

Annexure A

COMPARITIVE ANALYSIS

Criteria	Scope of Items (2)	Comparison Depth(2.5)	Use of Evidence (2)	Critical Thinking (2)	Clarity of Presentation (1)	Total(10)
Weightage	20%	25%	20%	20%	10%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Code of Conduct:

*I _C.Venkata Deepan (192425022) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION:

		Comparative analysis			
Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Scope of Items	20%	Selects highly relevant items with clear scope and justification	Selects appropriate items with minor gaps	Selection is limited or partially relevant	Items are unclear or not relevant
Comparison Depth	25%	Provides detailed and comprehensive comparison across multiple aspects	Provides adequate comparison with some depth	Limited comparison with few aspects	Very minimal or no comparison
Use of Evidence	20%	Supports comparison with accurate data, examples, or references	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Critical Thinking	20%	Demonstrates strong critical thinking with clear interpretation and reasoning	Shows reasonable interpretation with minor gaps	Basic interpretation; lacks depth	No meaningful interpretation
Clarity of Presentation	15%	Well-organized, clear, and logically structured presentation	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Annexure A

Q1 . Dataset: Assessment Tool 3: Comparative Analysis (25% → 5 Marks)

Time duration :45 Minutes
Assessment Tool 3 (Low)

Q1 . Dataset: Customer Data

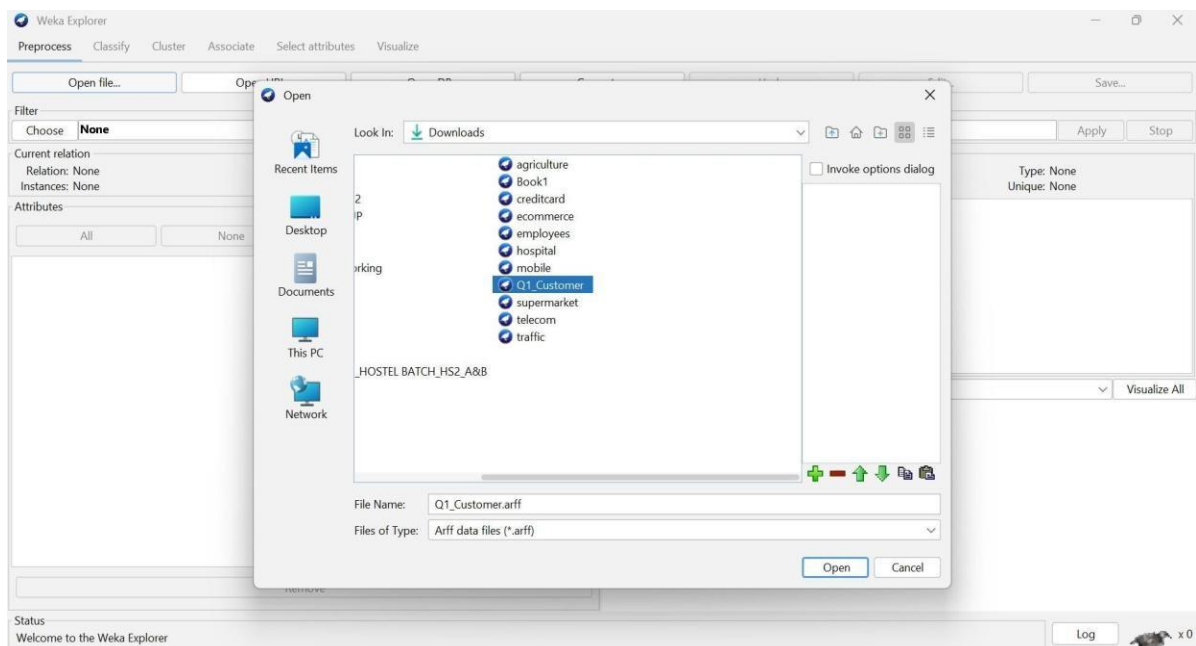
Customer Income (₹) Spending Score

C1	30000	60
C2	60000	40
C3	50000	70
C4	28000	65
C5	65000	35

Question:

Compare K-means and hierarchical clustering using this dataset and explain differences in performance and scalability. (BL3 – 1 Mark)

OUTPUT:



Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Cluster

Choose **HierarchicalClusterer** -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance -R first-last"

Cluster mode

☒ Use training set

☐ Supplied test set

☐ Percentage split

☐ Classes to clusters evaluation

(Num) SpendingScore

☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

06:38:09 - SimpleKMeans

06:38:50 - HierarchicalClusterer

Clusterer output

Initial starting points (random):

Cluster 0: C4,28000,65

Cluster 1: C2,60000,40

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data	Cluster#	0	1
	(5.0)	(3.0)	(2.0)	
Customer				
Income	46600	36000	62500	
SpendingScore	54	65	37.5	

Time taken to build model (full training data) : 0.02 seconds

=== Model and evaluation on training set ===

Clustered Instances

0	3 (60%)
1	2 (40%)

Status OK Log x0

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Cluster

Choose **HierarchicalClusterer** -N 2 -L SINGLE -P -A "weka.core.EuclideanDistance -R first-last"

Cluster mode

☒ Use training set

☐ Supplied test set

☐ Percentage split

☐ Classes to clusters evaluation

(Num) SpendingScore

☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)

06:38:09 - SimpleKMeans

06:38:50 - HierarchicalClusterer

Weka Clusterer Visualize: 06:38:09 - SimpleKMeans (customer_clustering)

X: Instance_number (Num)

Y: Customer (Nom)

Colour: Cluster (Nom)

Reset Clear Open Save

Jitter

Plot: customer_clustering_clustered

Class colour

cluster0 cluster1

Status OK Log x0

Weka Explorer

Preprocess Classify Cluster Associate Select attributes Visualize

Open file...

Open

Look In: Downloads

Recent Items

2

p

Desktop

Documents

Working

This PC

Network

Invoke options dialog

Type: None

Unique: None

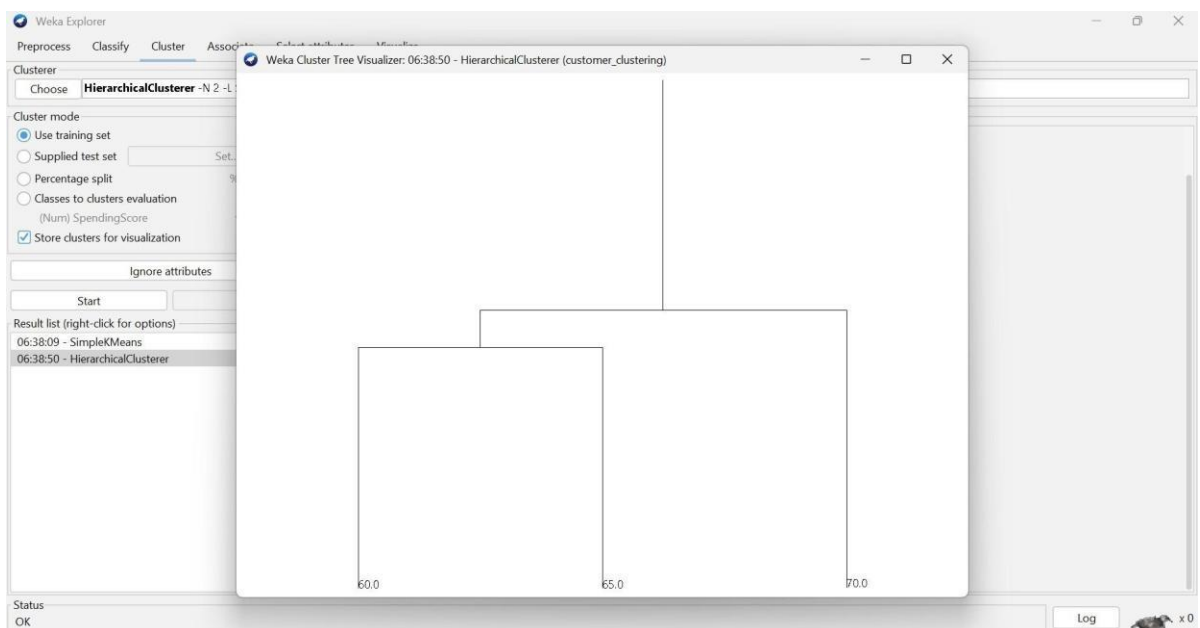
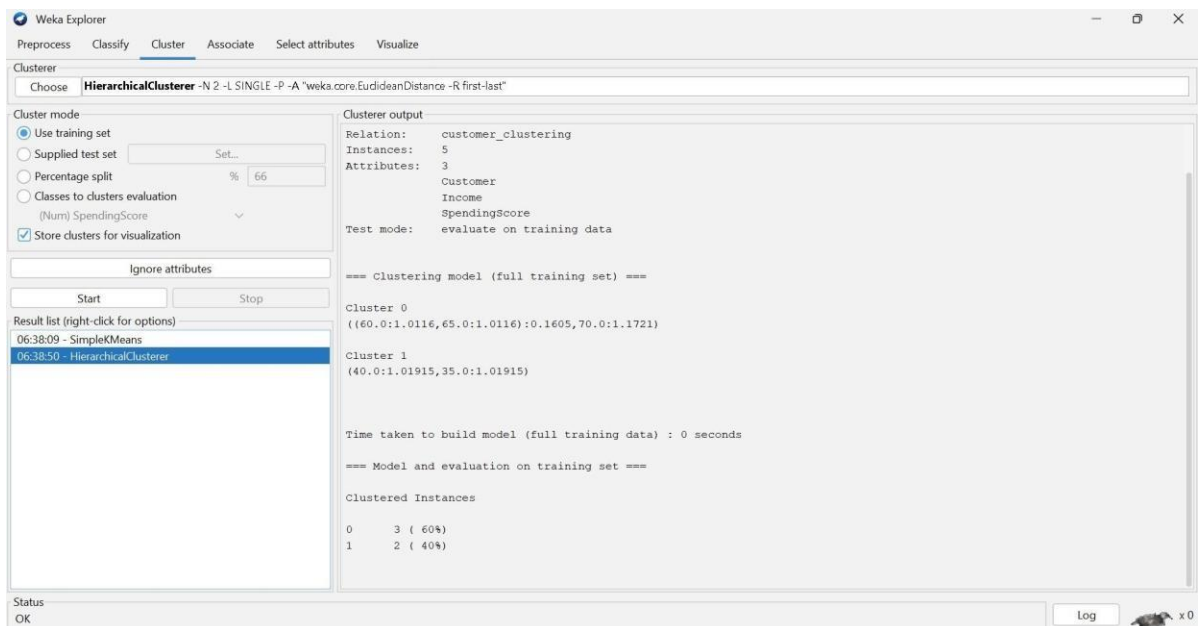
Visualize All

File Name: Q1_Customer.arff

Files of Type: Arff data files (*.arff)

Open Cancel

Status Welcome to the Weka Explorer Log x0



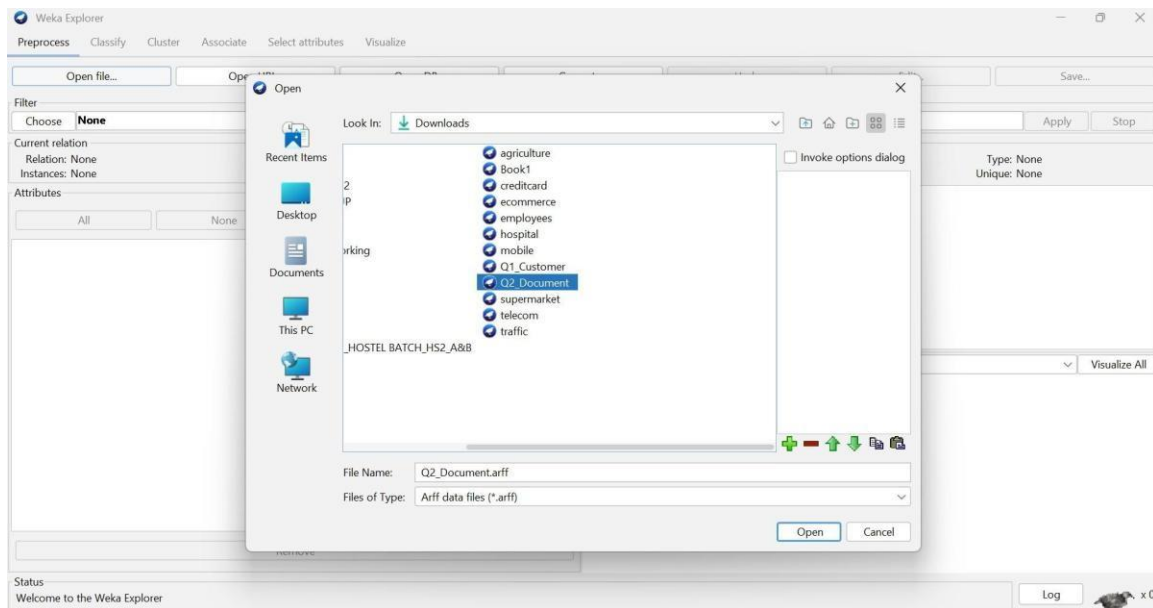
Q2. Dataset: Document Data

Document Word1 Word2 Word3

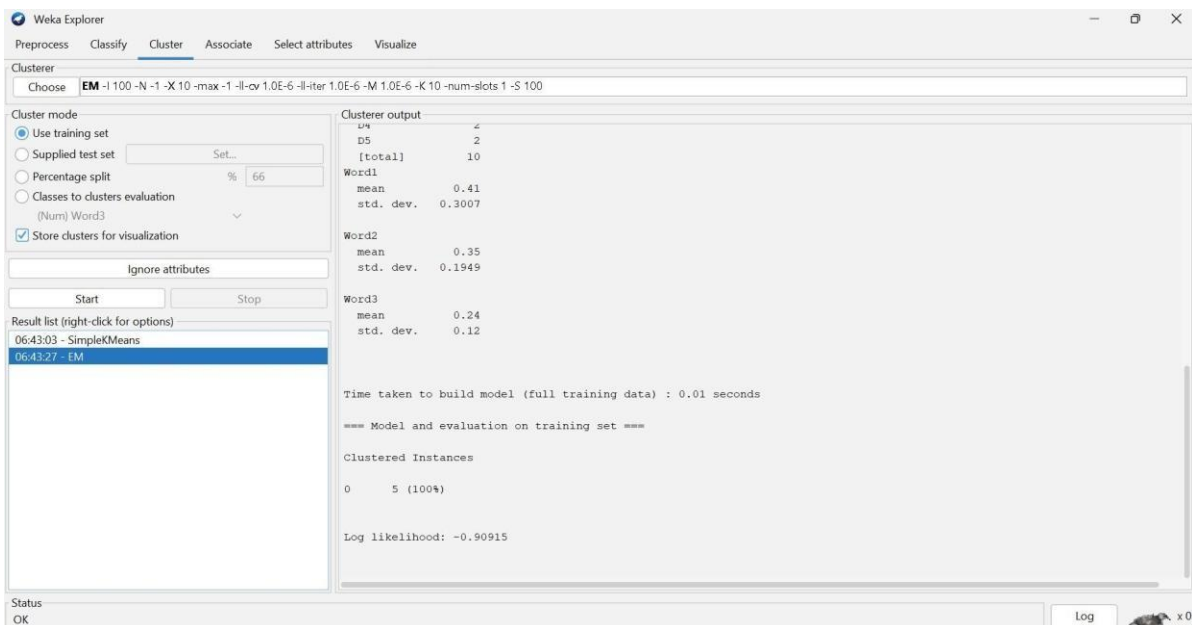
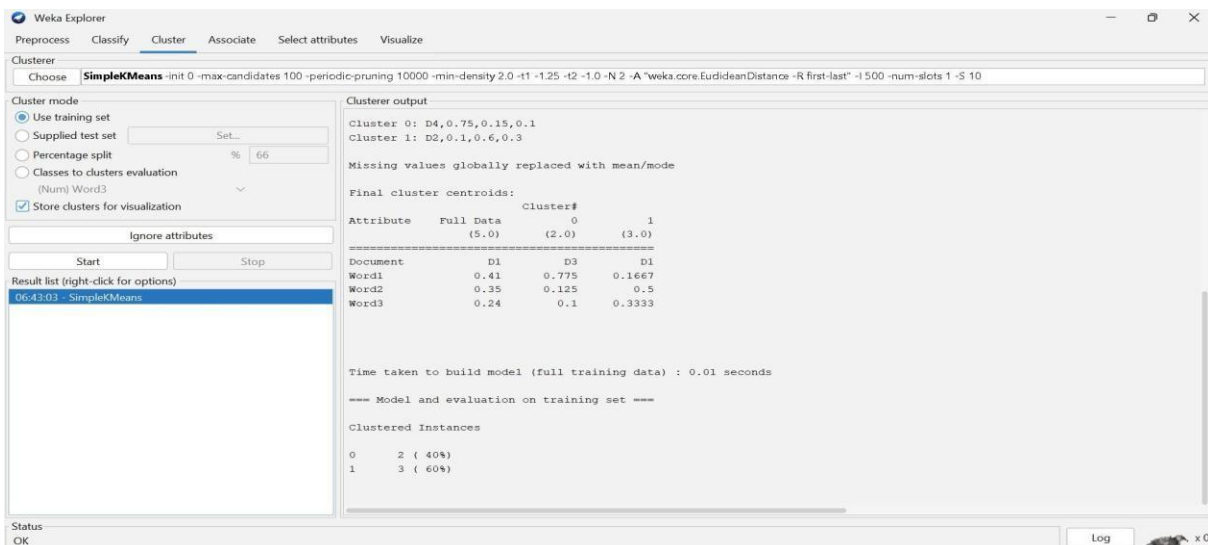
D1	0.2	0.5	0.3
D2	0.1	0.6	0.3
D3	0.8	0.1	0.1
D4	0.75	0.15	0.1
D5	0.2	0.4	0.4

Question:

Compare K-means and EM algorithms in handling overlapping clusters for the given dataset.
(BL4 – 1 Mark)



OUTPUT:



Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer: Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode:
☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation (Num) Word3
☒ Store clusters for visualization
 Ignore attributes
 Start Stop

Result list (right-click for options):
 06:43:03 - SimpleKMeans

Cluster output:

```
Cluster 0: D4,0.75,0.15,0.1
Cluster 1: D2,0.1,0.6,0.3

Missing values globally replaced with mean/mode

Final cluster centroids:
Attribute Full Data Cluster#
(5.0) (2.0) (3.0)
=====
Document D1 D3 D1
Word1 0.41 0.775 0.1667
Word2 0.35 0.125 0.5
Word3 0.24 0.1 0.3333

Time taken to build model (full training data) : 0.01 seconds

=== Model and evaluation on training set ===

Clustered Instances
0 2 ( 40%)
1 3 ( 60%)
```

Status OK Log x0

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer: Choose **EM** -I 100 -N -1 -X 10 -max -1 -ll-cv 1.0E-6 -ll-iter 1.0E-6 -M 1.0E-6 -K 10 -num-slots 1 -S 100

Cluster mode:
☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation (Num) Word3
☒ Store clusters for visualization
 Ignore attributes
 Start Stop

Result list (right-click for options):
 06:43:03 - SimpleKMeans
 06:43:27 - EM

Cluster output:

```

D5 2
D5 2
[total] 10
Word1
mean 0.41
std. dev. 0.3007

Word2
mean 0.35
std. dev. 0.1949

Word3
mean 0.24
std. dev. 0.12

Time taken to build model (full training data) : 0.01 seconds

=== Model and evaluation on training set ===

Clustered Instances
0 5 (100%)

Log likelihood: -0.90915
```

Status OK Log x0

Weka Explorer

Preprocess Classify **Cluster** Associate Select attributes Visualize

Clusterer: Choose **EM** -I 100 -N -1 -X 10 -max -1

Cluster mode:
☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation (Num) Word3
☒ Store clusters for visualization
 Ignore attributes
 Start Stop

Result list (right-click for options):
 06:43:03 - SimpleKMeans
 06:43:27 - EM

Weka Clusterer Visualize: 06:43:27 - EM (document_clustering)

X: Instance_number (Num) Y: Document (Nom)
 Colour: Cluster (Nom) Select Instance
 Reset Clear Open Save Jitter

Plot: document_clustering_clustered

Class colour cluster0

Status OK Log x0

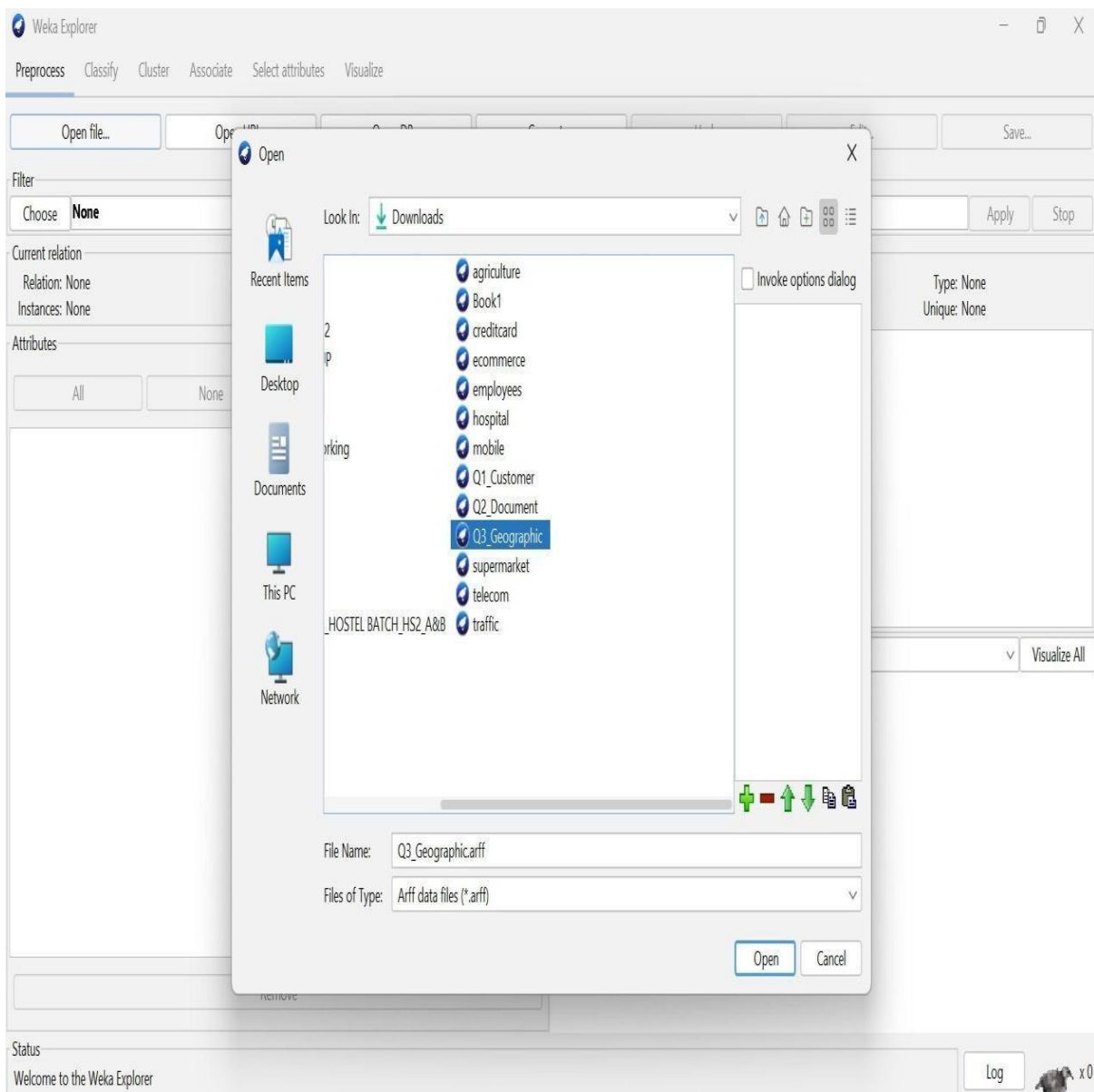
Q3. Dataset: Geographic Data

Point Latitude Longitude

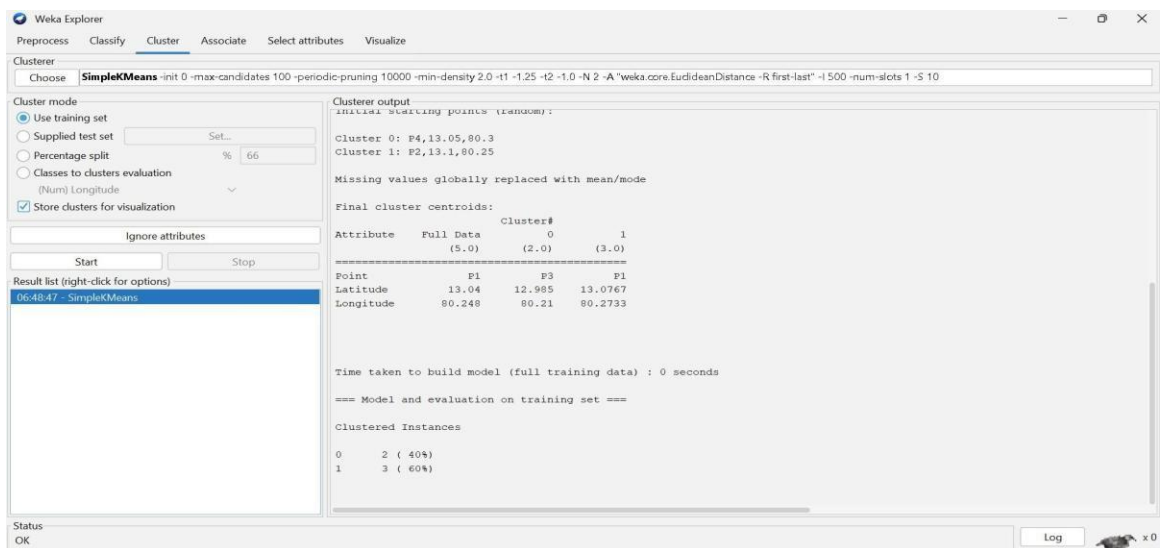
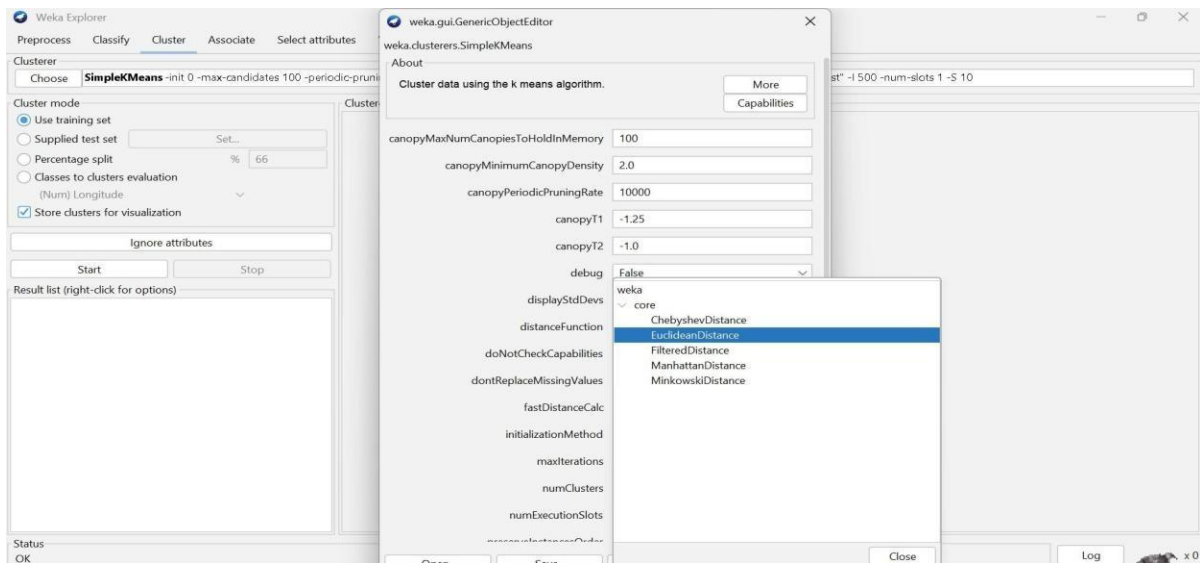
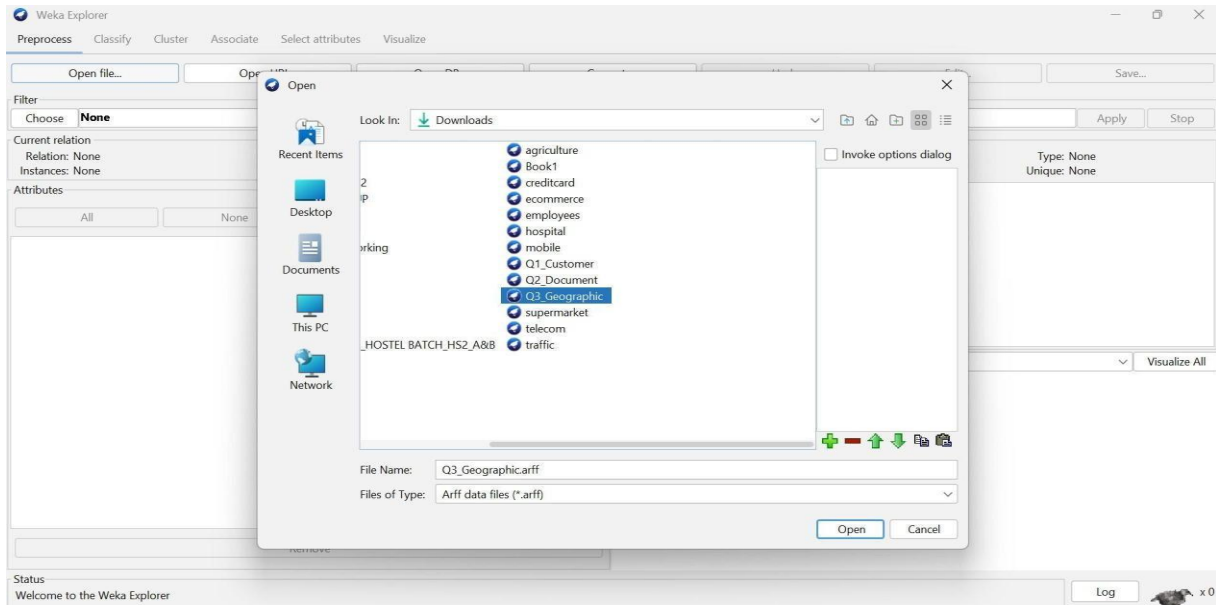
P1	13.08	80.27
P2	13.10	80.25
P3	12.97	80.22
P4	13.05	80.30
P5	13.00	80.20

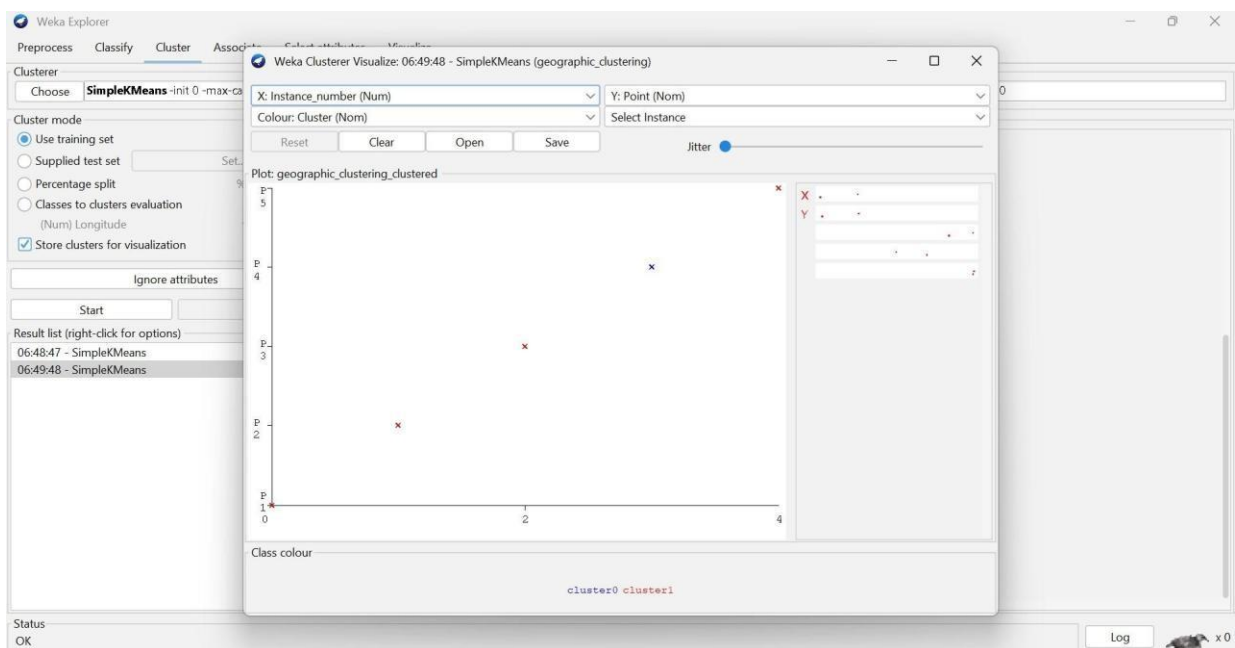
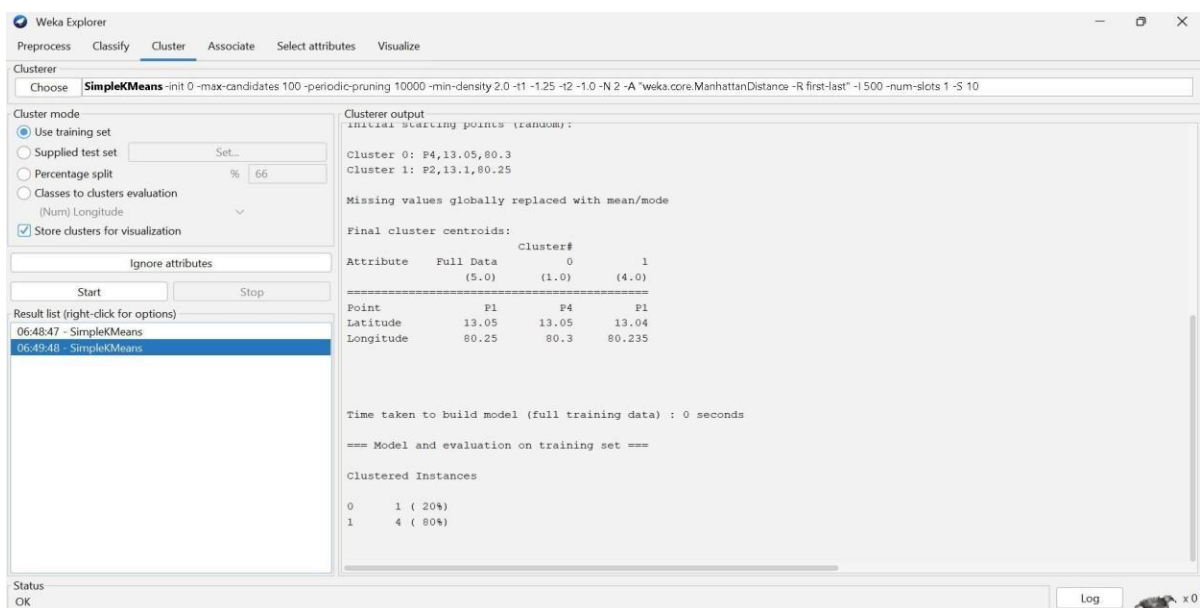
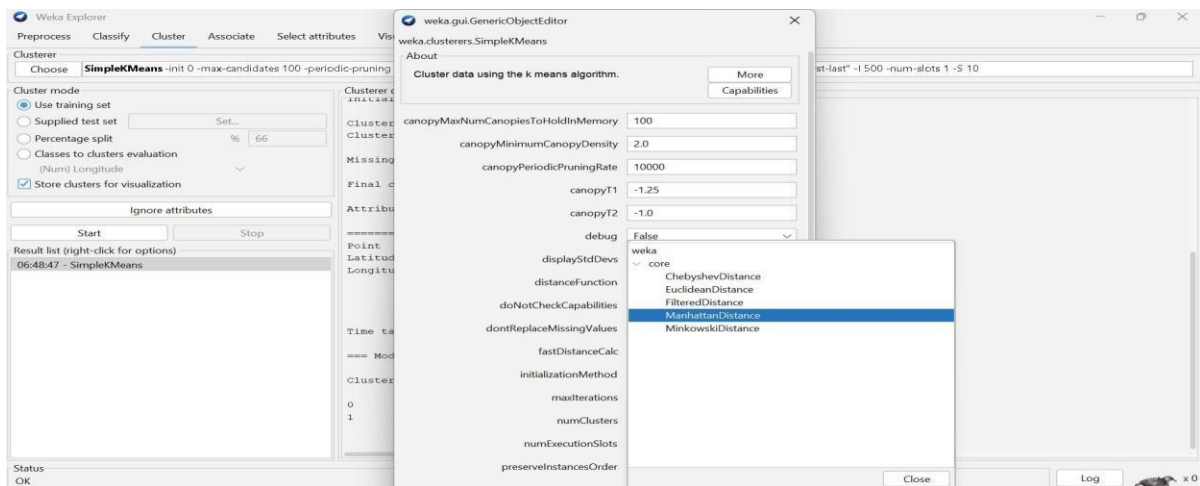
Question:

Compare Euclidean and Manhattan distance metrics and analyze their impact on clustering results. (BL4 – 1 Mark)



OUTPUT:





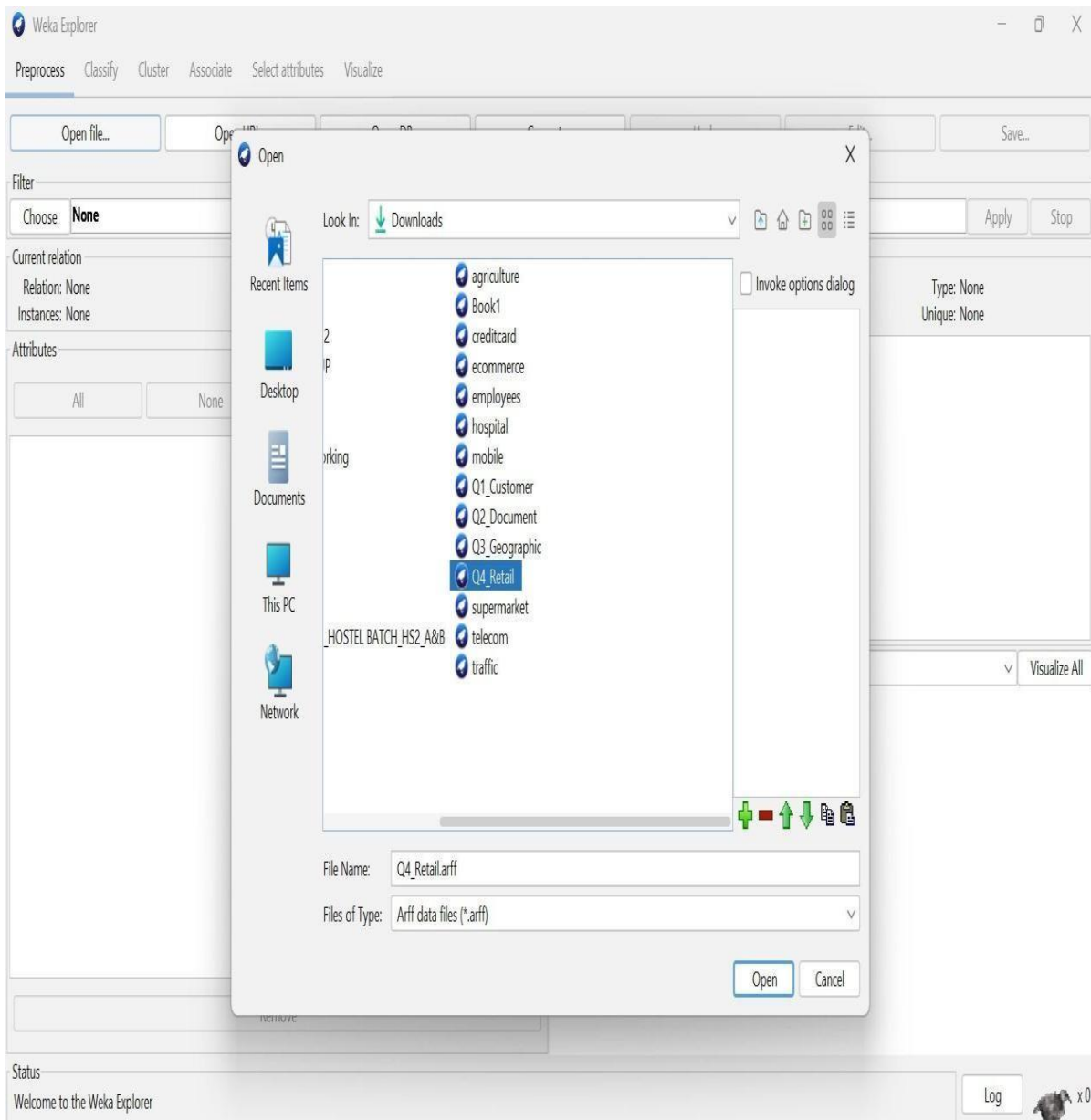
Q4. Dataset: Retail Products

Product Sales (₹) Frequency

P1	50000	20
P2	30000	10
P3	52000	22
P4	25000	8
P5	32000	12

Question:

Compare agglomerative and divisive clustering approaches and explain their suitability for this dataset. (BL3 – 1 Mark)



OUTPUT:

The screenshot shows the Weka Explorer interface with the 'Cluster' tab selected. The 'HierarchicalClusterer' is chosen, and the 'Cluster mode' is set to 'Use training set'. The 'Clusterer output' pane displays the following information:

Relation: retail_products
Instances: 5
Attributes: 3
Product
Sales
Frequency
Test mode: evaluate on training data

=== Clustering model (full training set) ===

Cluster 0
(20.0:1.01286,22.0:1.01286)

Cluster 1
((10.0:1.01286,12.0:1.01286):0.01412,8.0:1.02699)

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0	2 (40%)
1	3 (60%)

The 'Result list' on the left shows the operation '06:53:39 - HierarchicalClusterer' as the most recent entry.

Weka Explorer

Preprocess **Classify** **Cluster** Associate Select attributes Visualize

Clusterer
Choose **MakeDensityBasedClusterer** -M 1.0E-6 -W weka.clusterers.SimpleKMeans -- -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -n

Cluster mode
☒ Use training set
☐ Supplied test set Set...
☐ Percentage split % 66
☐ Classes to clusters evaluation (Num) Frequency
☒ Store clusters for visualization

Ignore attributes

Start Stop

Result list (right-click for options)
 06:53:39 - HierarchicalClusterer
06:54:50 - MakeDensityBasedClusterer

Cluster output

```

Discrete Estimator. Counts = 1 2 1 2 2 (Total = 8)
Attribute: Sales
Normal Distribution. Mean = 29000 StdDev = 2943.9203
Attribute: Frequency
Normal Distribution. Mean = 10 StdDev = 1.633

Cluster: 1 Prior probability: 0.4286

Attribute: Product
Discrete Estimator. Counts = 2 1 2 1 1 (Total = 7)
Attribute: Sales
Normal Distribution. Mean = 51000 StdDev = 1000
Attribute: Frequency
Normal Distribution. Mean = 21 StdDev = 1

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0      3 ( 60%)
1      2 ( 40%)

Log likelihood: -12.6953
  
```

Status
OK

Log x 0

Weka Explorer

Preprocess **Classify** **Cluster** Associate Select attributes Visualize

Clusterer
Choose **MakeDensityBasedClusterer**

Cluster mode
☒ Use training set
☐ Supplied test set Set...
☐ Percentage split %
☐ Classes to clusters evaluation (Num) Frequency
☒ Store clusters for visualization

Ignore attributes

Start

Result list (right-click for options)
 06:53:39 - HierarchicalClusterer
06:54:50 - MakeDensityBasedClusterer

Weka Clusterer Visualize: 06:54:50 - MakeDensityBasedClusterer (retail_products)

X: Instance_number (Num) Y: Product (Nom)
 Colour: Cluster (Nom) Select Instance
 Reset Clear Open Save Jitter

Plot: retail_products_clustered

Class colour
cluster0 cluster1

Status
OK

Log x 0

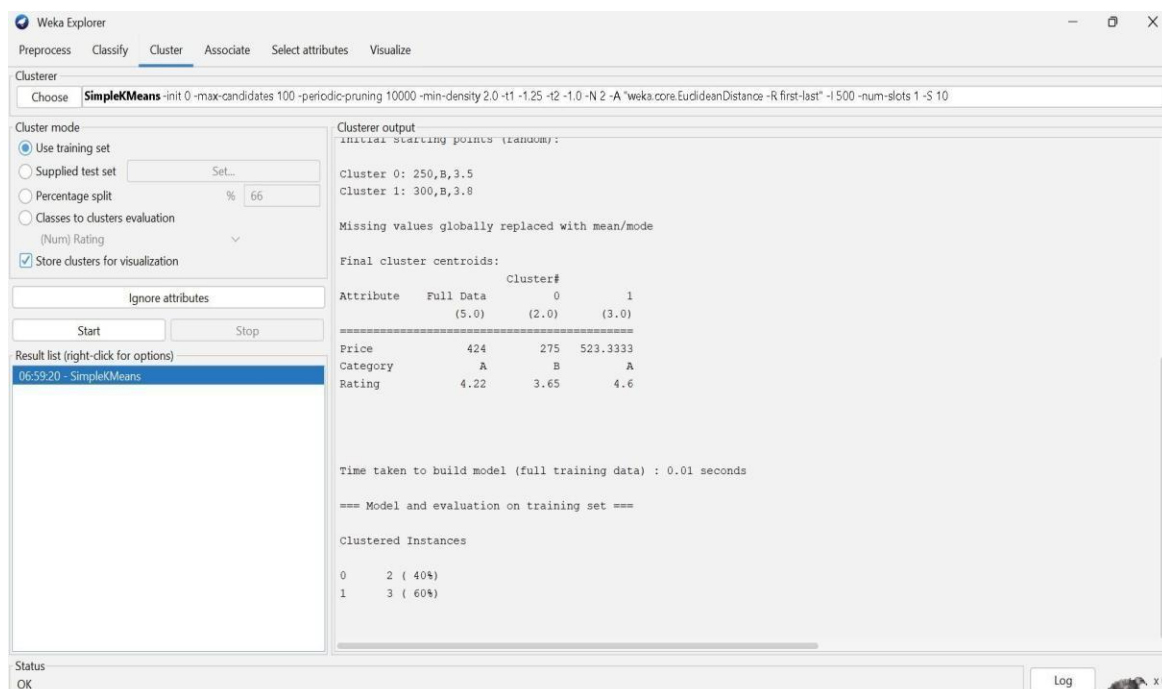
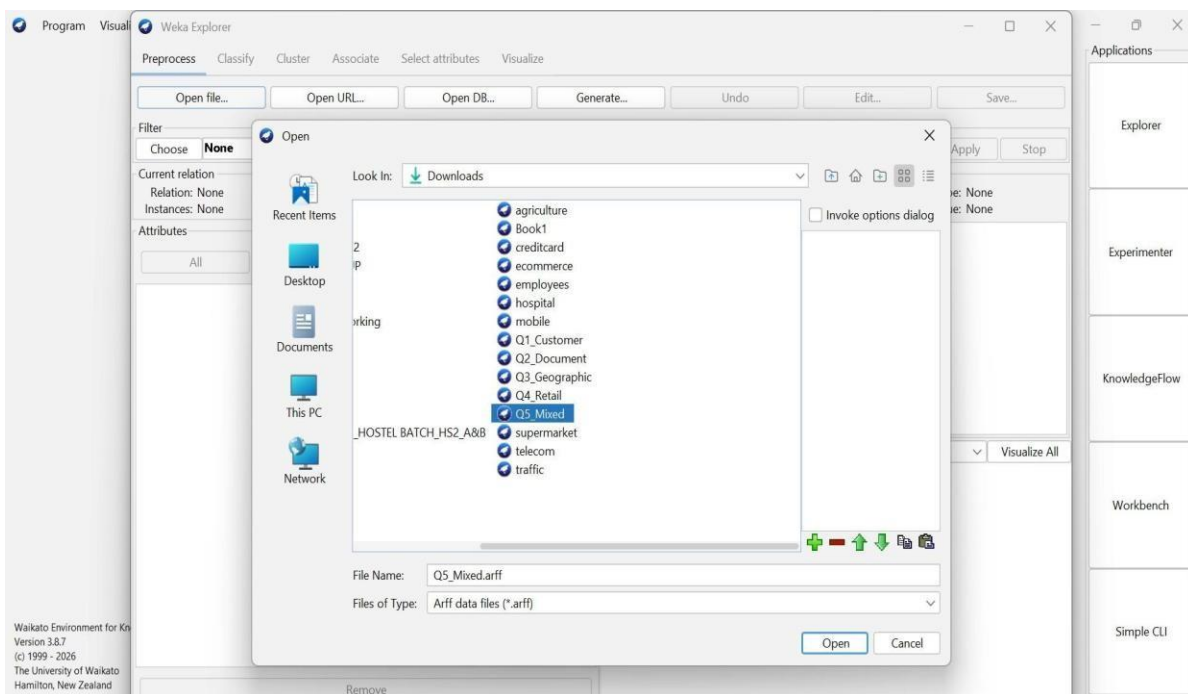
Q5 . Dataset: Mixed Attributes

Item Price (₹) Category Rating

I1	500	A	4.5
I2	300	B	3.8
I3	550	A	4.7
I4	250	B	3.5
I5	520	A	4.6

Question:

Compare centroid-based and density-based clustering techniques and analyze their effectiveness on mixed data types. (BL4 – 1 Mark)



OUTPUT:

